

AYUSHMAN SAHOO

Software Development Engineer | Machine Learning Engineer

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EDUCATION

B.Tech in Computer Science Engineering (AI & ML)

Aug 2024 – May 2028

VIT-AP University, Amaravati, Andhra Pradesh

TECHNICAL SKILLS

Programming Python, Java, TypeScript, JavaScript, C, SQL

Software Engineering Data Structures & Algorithms, OOP, REST APIs, API Integration, Git, GitHub

Frontend React.js, Next.js, Three.js, HTML, CSS, Tailwind CSS

Backend Node.js, Express.js, FastAPI, Flask

Databases PostgreSQL, MongoDB

Machine Learning Supervised Learning, Unsupervised Learning, Deep Learning, Reinforcement Learning

AI / Generative AI LLMs, RAG, AI Agents, Prompt Engineering, LangChain

ML Libraries PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas

DevOps & Tools Docker, Linux, Postman, CI/CD

PROJECTS

Telco-RCA 2025

[GitHub](#)

Python · FastAPI · React · Groq (Llama 3.3 70B) · Docker · Vite · Tailwind CSS

- Built an AI-driven root cause analysis system using graph reasoning and autonomous fault isolation for 5G network outages, deployed live on Hugging Face Spaces.
- Designed a 4-layer hierarchical 5G knowledge graph supporting up to 1,000 nodes across multiple difficulty levels, with causal shuffling and anti-loop safeguards.
- Integrated Llama 3.3 70B through the Groq API for sub-second causal inference using a Functional Cognitive Loop (Trace → Check → Diagnose) grounded on telemetry and alarm feeds.
- Engineered evaluation analytics for MTTR, diagnostic accuracy, and reasoning efficiency; containerized the application with Docker and served it through FastAPI and Uvicorn.

OceanEmbed — Satellite Subsurface Ocean Temperature Reconstruction 2026

[GitHub](#)

Team Project · Python · PyTorch · FastAPI · TypeScript · React · Docker

- Contributed to a team-built deep learning system for reconstructing 3D subsurface ocean temperature fields from satellite-derived surface ocean variables across the North Indian Ocean.
- Developed and integrated FastAPI REST endpoints for geospatial data processing, model inference, temperature profiles, and ARGO-based validation.
- Implemented the TypeScript/React visualization layer with Vite, API proxying, WebGL heatmaps, and safe-mode fallbacks for reliable demonstrations.
- Containerized frontend and ML backend services using Docker Compose, with persistent data, model, and output volumes for reproducible execution.

Kairo 2025 — Ongoing

[GitHub](#)

React.js · Next.js · TypeScript · Tailwind CSS · Node.js

- Building a full-stack event discovery platform using Next.js, TypeScript, and Node.js to curate tech events, hackathons, concerts, and meetups; deployed on Vercel.
- Designed RESTful APIs and a structured data layer indexing 500+ active events across 50+ cities for event discovery and ranking.
- Implemented authentication, personalized event feeds, and save/bookmark functionality using reusable React components and responsive Tailwind CSS interfaces.

CERTIFICATIONS

- Agentic AI — Oracle (2026)**
- JavaScript Certification — HackerRank (2026)**
- Java Programming Certification — CTTC, MSME Affiliated (2025)**
- Python Programming Certification — CTTC, MSME Affiliated (2025)**
- C Programming Certification — CTTC, MSME Affiliated (2024)**